

INTRODUCTION TO CIRCUIT DESIGN AND SPICE SIMULATIONS

Why do we need SPICE simulations?

SPICE simulations verify circuit behavior and performance before implementation. In **circuit design**, logic gates like NAND, NOR, and NOT are built using NMOS and PMOS transistors.

For example, in an **inverter**:

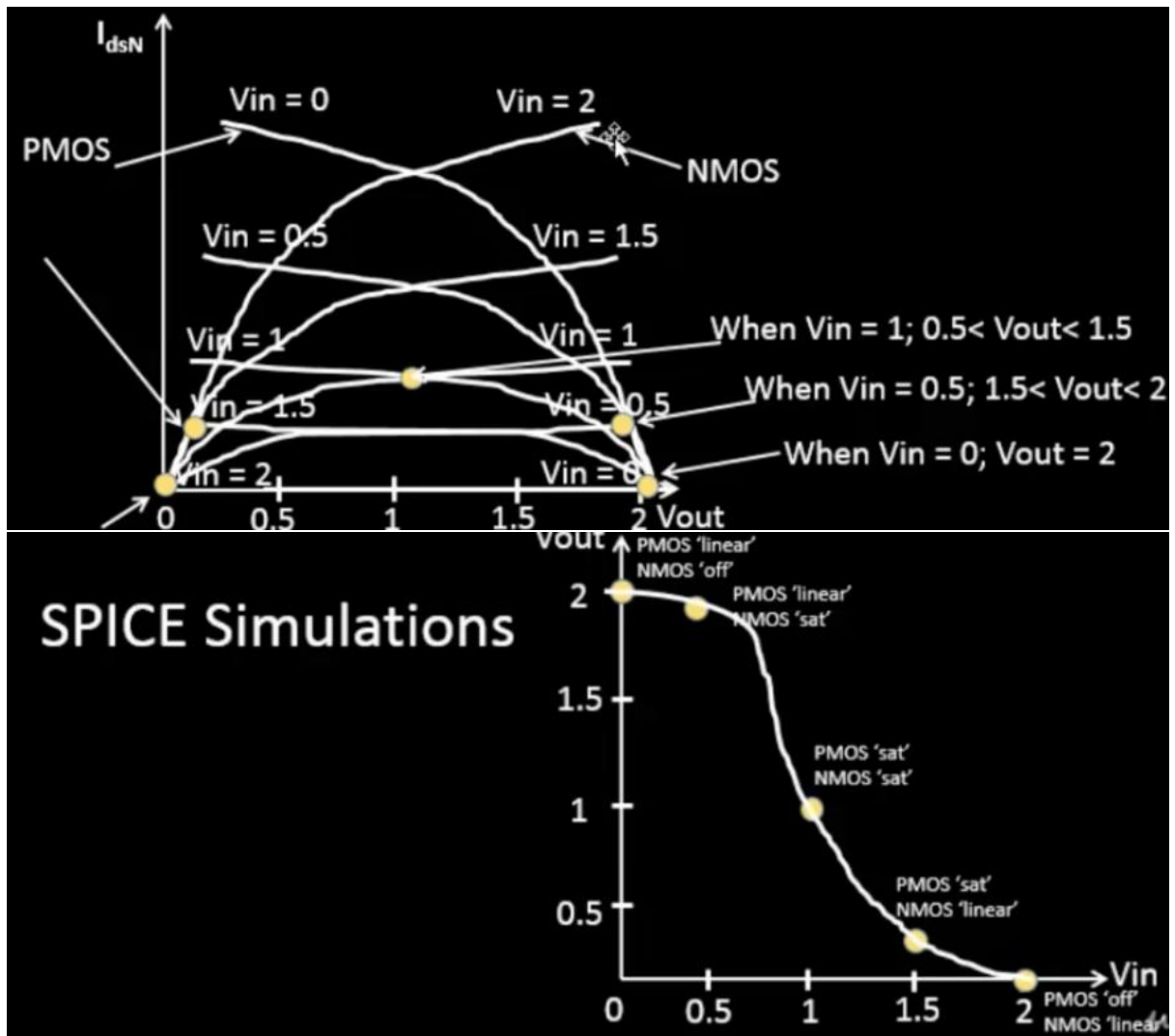
- PMOS (pull-up) and NMOS (pull-down) drains connect to the **output** with a capacitive load.
- Both **gates** connect to the **input**, PMOS source to **VDD**, and NMOS source to **GND**.

SPICE ensures the circuit meets functionality, timing, and power requirements.

SPICE Simulation:

SPICE simulations let us test and fine-tune circuits virtually before building them. For example, if we provide an **input waveform** to a circuit, SPICE generates the **output waveform**, showing how the circuit behaves.

These waveforms help us measure things like **delay**. By tweaking the design parameters, such as the **width (W)** and **length (L)** of transistors, we can control the **current flow**, which in turn affects the delay. This makes it easier to optimize the circuit for better performance.



Example

Power Aware CTS

Delay Table for CBUF '1'							Delay Table for CBUF '2'								
Input slew	Output Load						Input slew	Output Load							
	10fF	30fF	50fF	70fF	90fF	110fF		10fF	30fF	50fF	70fF	90fF	110fF		
	20ps	x1	x2	x3	x4	x5		x6	20ps	y1	y2	y3	y4	y5	y6
	40ps	x7	x8	x9	x10	x11		x12	40ps	y7	y8	y9	y10	y11	y12
	60ps	x13	x14	x15	x16	x17		x18	60ps	y13	y14	y15	y16	y17	y18
80ps	x19	x20	x21	x22	x23	x24	80ps	y19	y20	y21	y22	y23	y24		

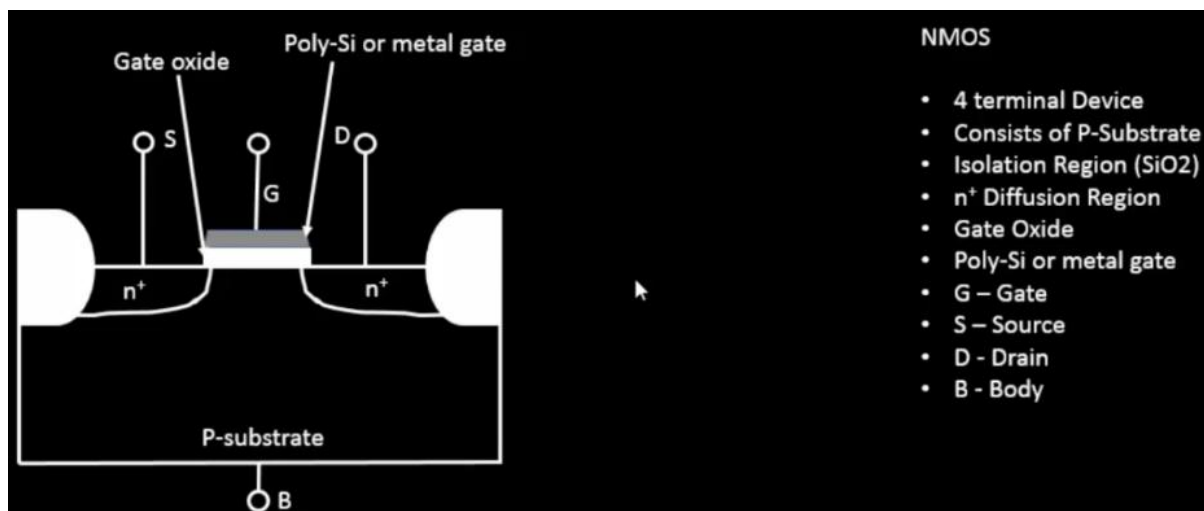
- "Did you know, where does the delay of a cell actually comes from?"
- "We have learnt about delay models, but are the models accurate?"
- "How do you verify, if what you are doing in static timing analysis, is correct?"

The difference between **CBUF1** and **CBUF2** comes from their **circuit design**, which might have different **drive strengths** or **W/L ratios** (transistor width-to-length).

The **delay table values** are obtained from **SPICE simulations**, which are also used to verify delays for the circuit design.

Introduction to Basic Element in Circuit Design - NMOS Transistor

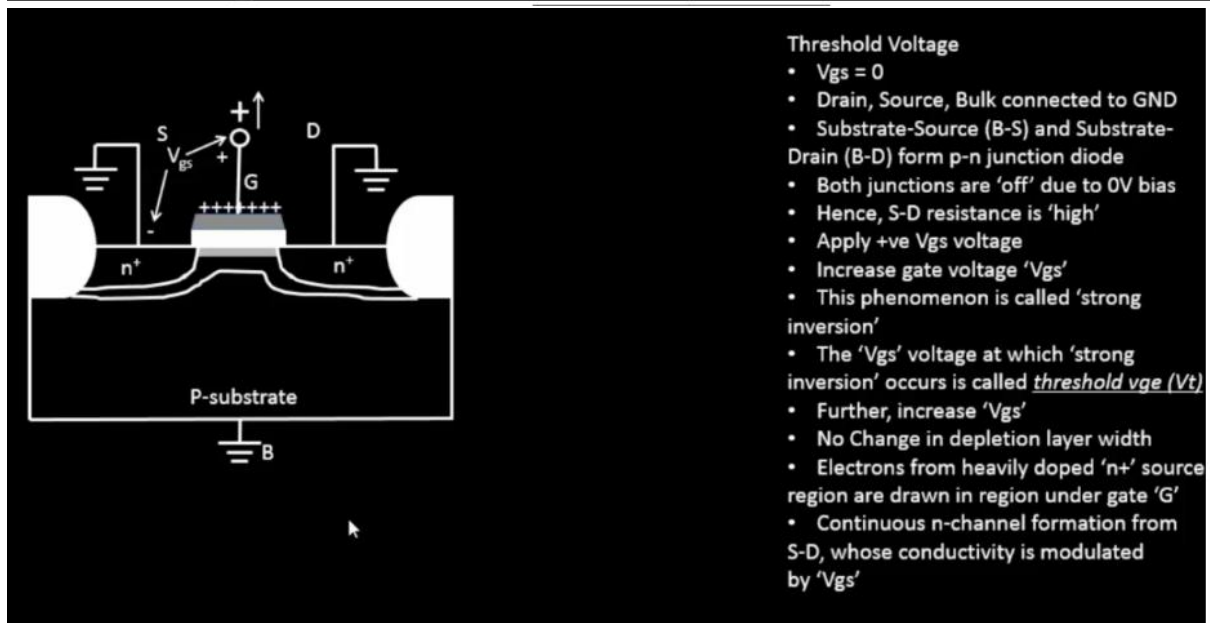
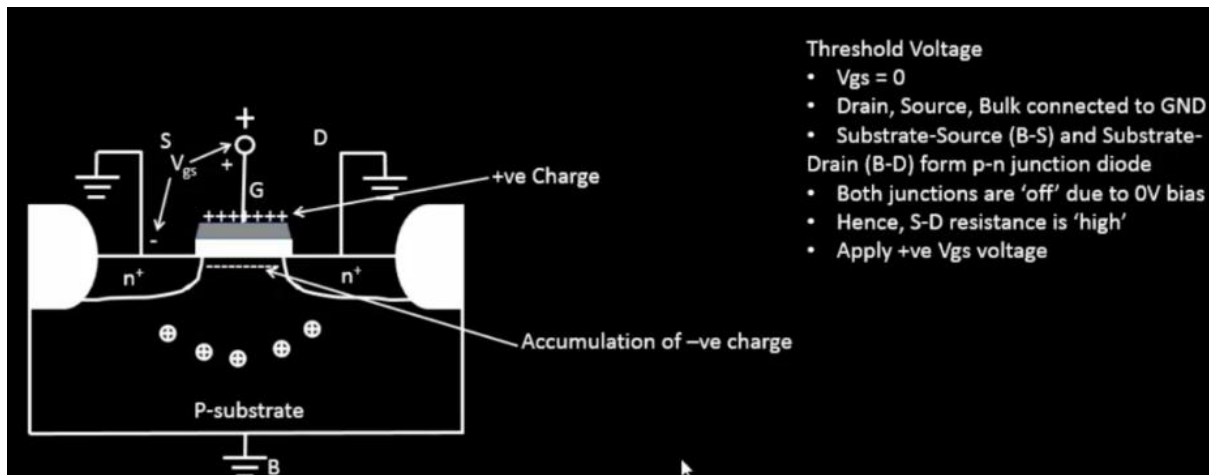
NMOS Transistor: An N-type Metal-Oxide-Semiconductor that conducts when a positive voltage is applied to its gate, enabling current flow from drain to source.



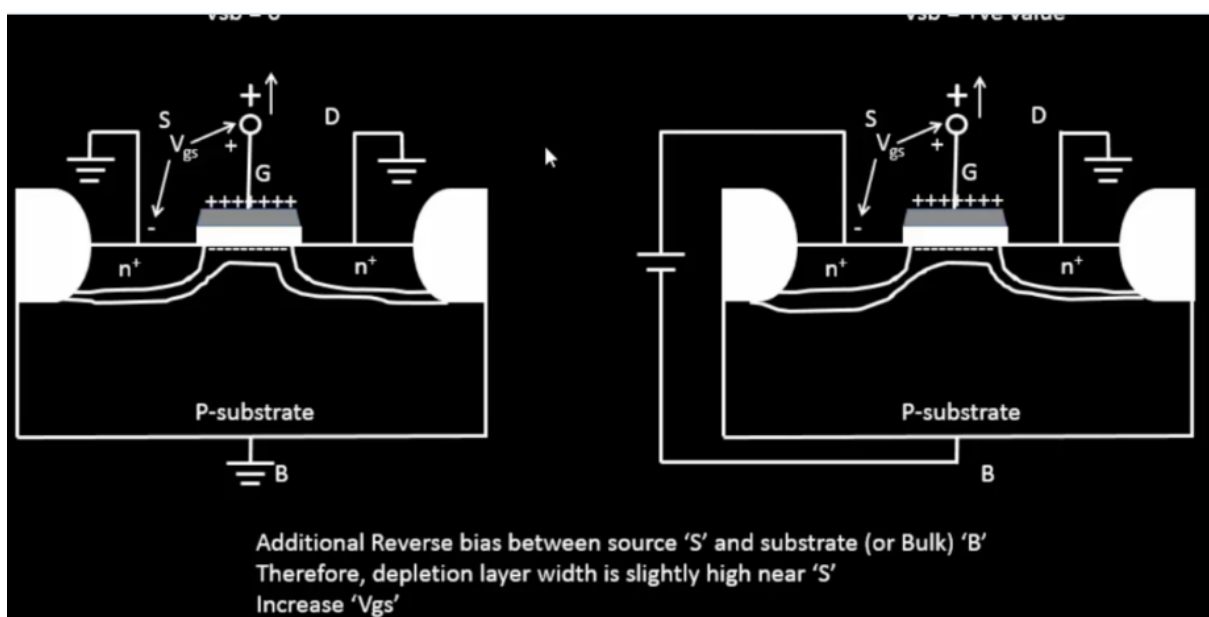
Threshold Voltage: A key parameter that drives SPICE simulations, representing the voltage at which a transistor switches from off to on.

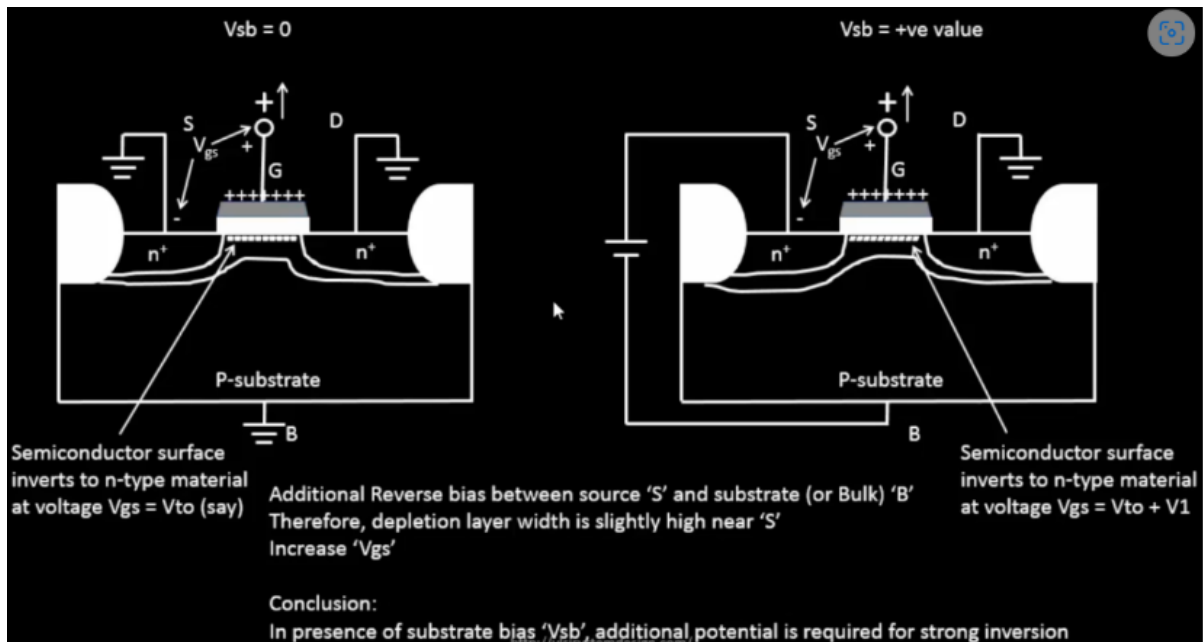
Strong Inversion: The condition where the transistor channel is fully formed, allowing significant current flow.

Threshold Voltage with Substrate Potential: The threshold voltage varies with changes in substrate potential due to the **body effect**.



Fourth Terminal Body





Threshold Voltage at $V_{sb} = 0$: The threshold voltage when the source-to-body voltage is zero.

Body Effect Coefficient and Fermi Potential: Constants provided by the foundry that form the transistor model and are fed into SPICE simulations for various calculations.

Threshold Voltage Equation:

$$V_t = V_{to} + \gamma (\sqrt{|-2\Phi_f + V_{sb}|} - \sqrt{|-2\Phi_f|})$$

Where

- V_{to} = Threshold voltage at $V_{sb} = 0$, and is a function of manufacturing process
- γ = body effect coefficient, expresses the impact of changes in body bias V_{sb} (Unit is $V^{0.5}$)
- Φ_f = Fermi Potential (Covered in Semiconductor Physics Slides)

$$\gamma = \frac{\sqrt{2qN_A\epsilon_{si}}}{C_{ox}}$$

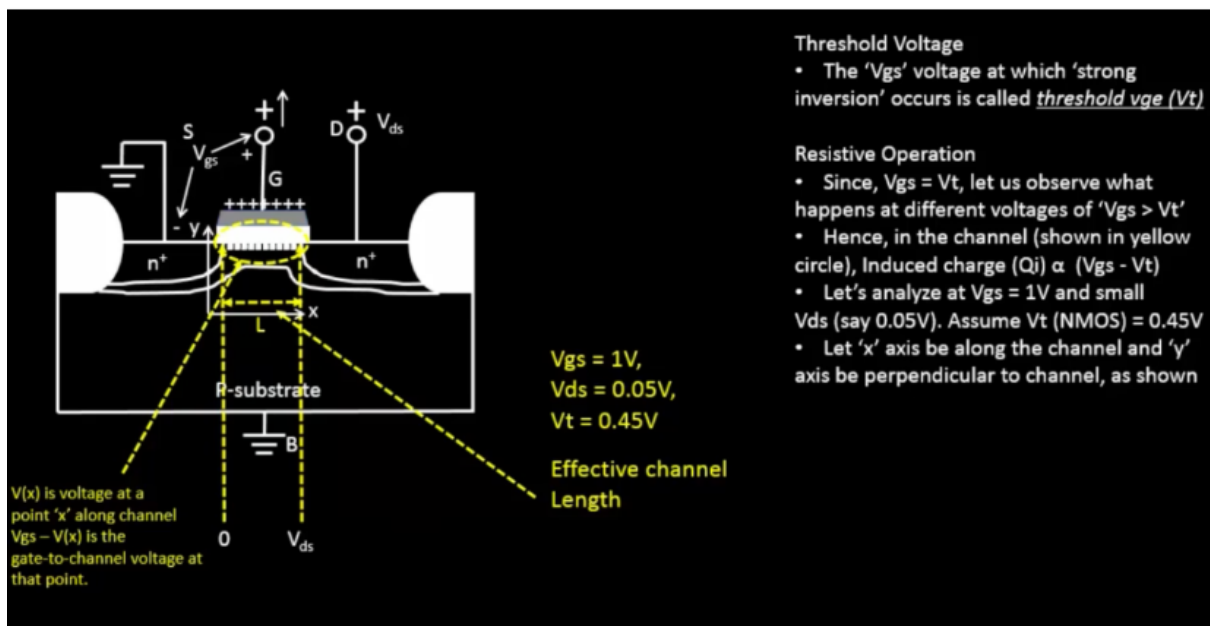
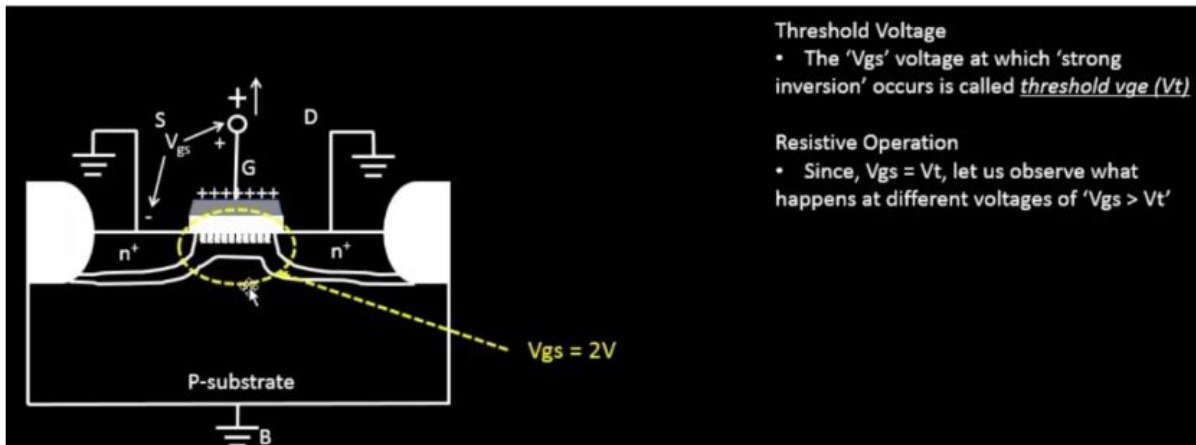
- ϵ_{si} = relative permittivity of silicon = 11.7
- N_A = doping concentration
- q = charge of the electron
- C_{ox} = oxide capacitance

$$\Phi_f = -\Phi_T \ln \frac{N_A}{n_i}$$

n_i = intrinsic doping parameter for the substrate

NMOS Resistive and Saturation Regions of Operation

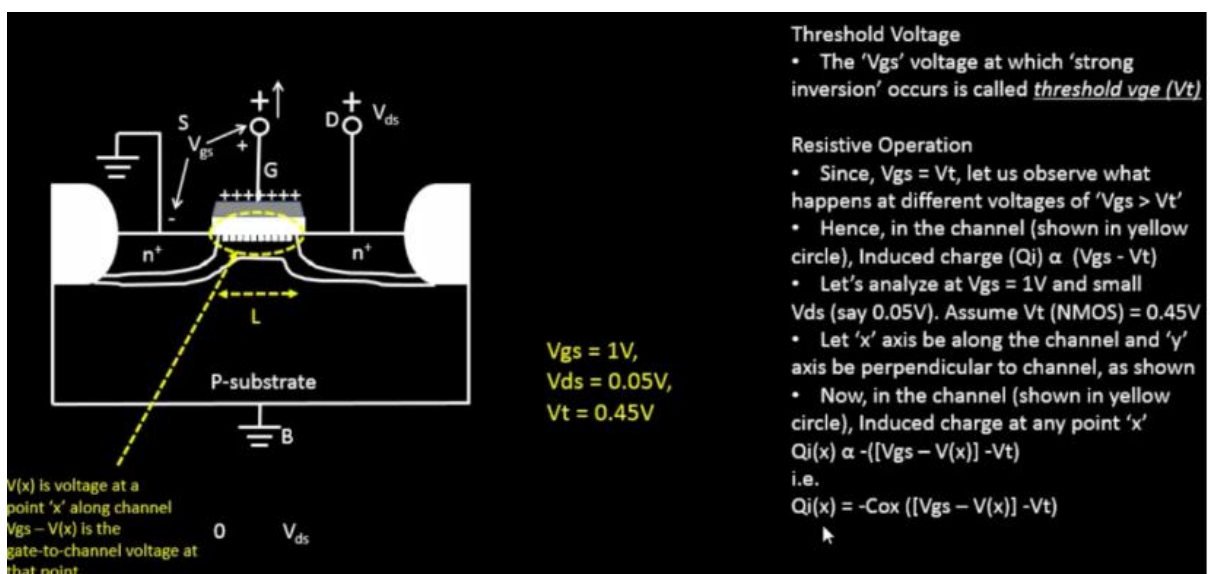
Resistive region of operation with small drain-source voltage

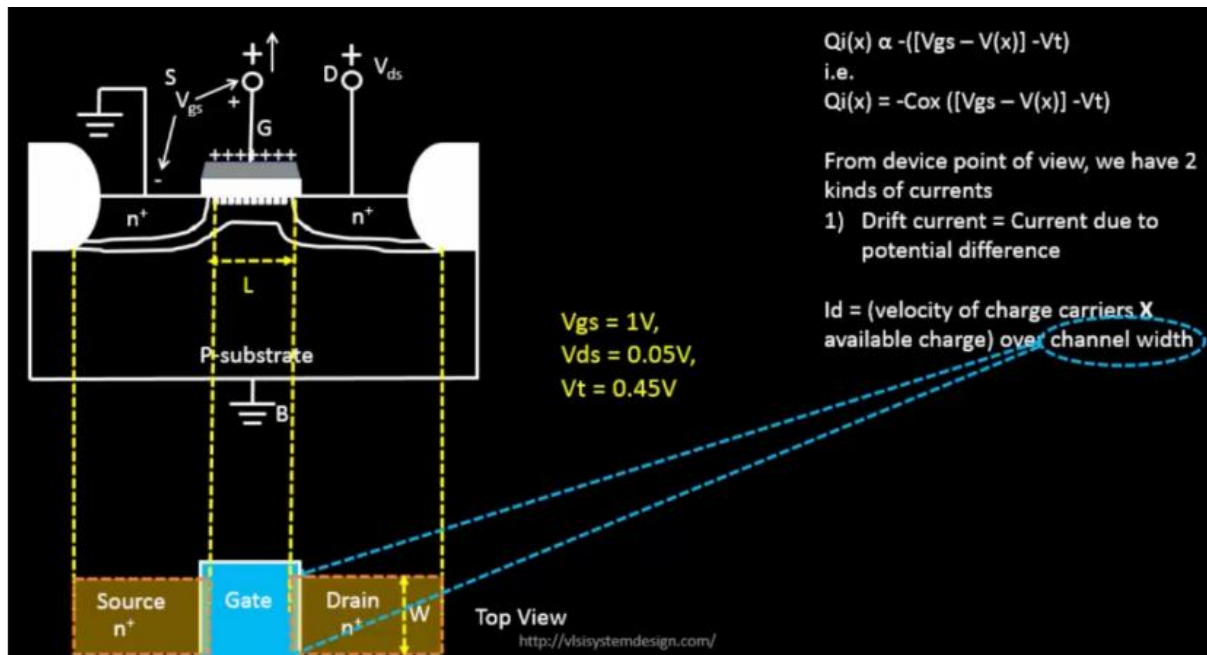


Drift Current: Current caused by a potential difference across the material.

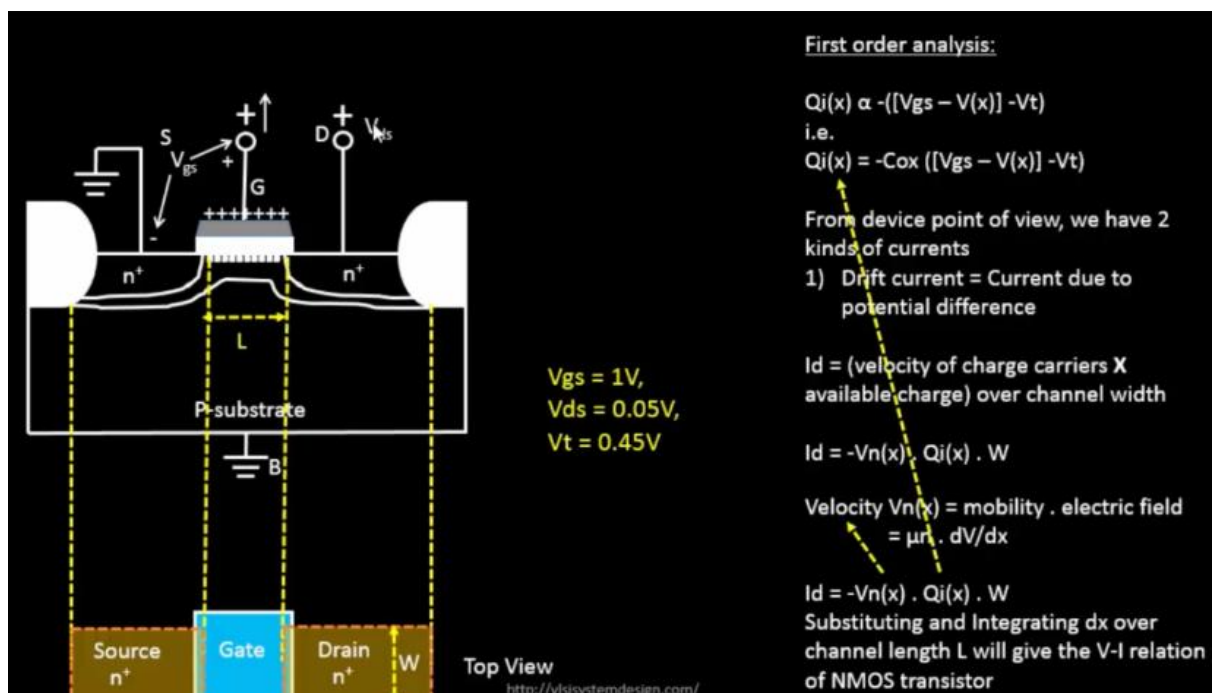
Diffusion Current: Current resulting from a difference in carrier concentration.

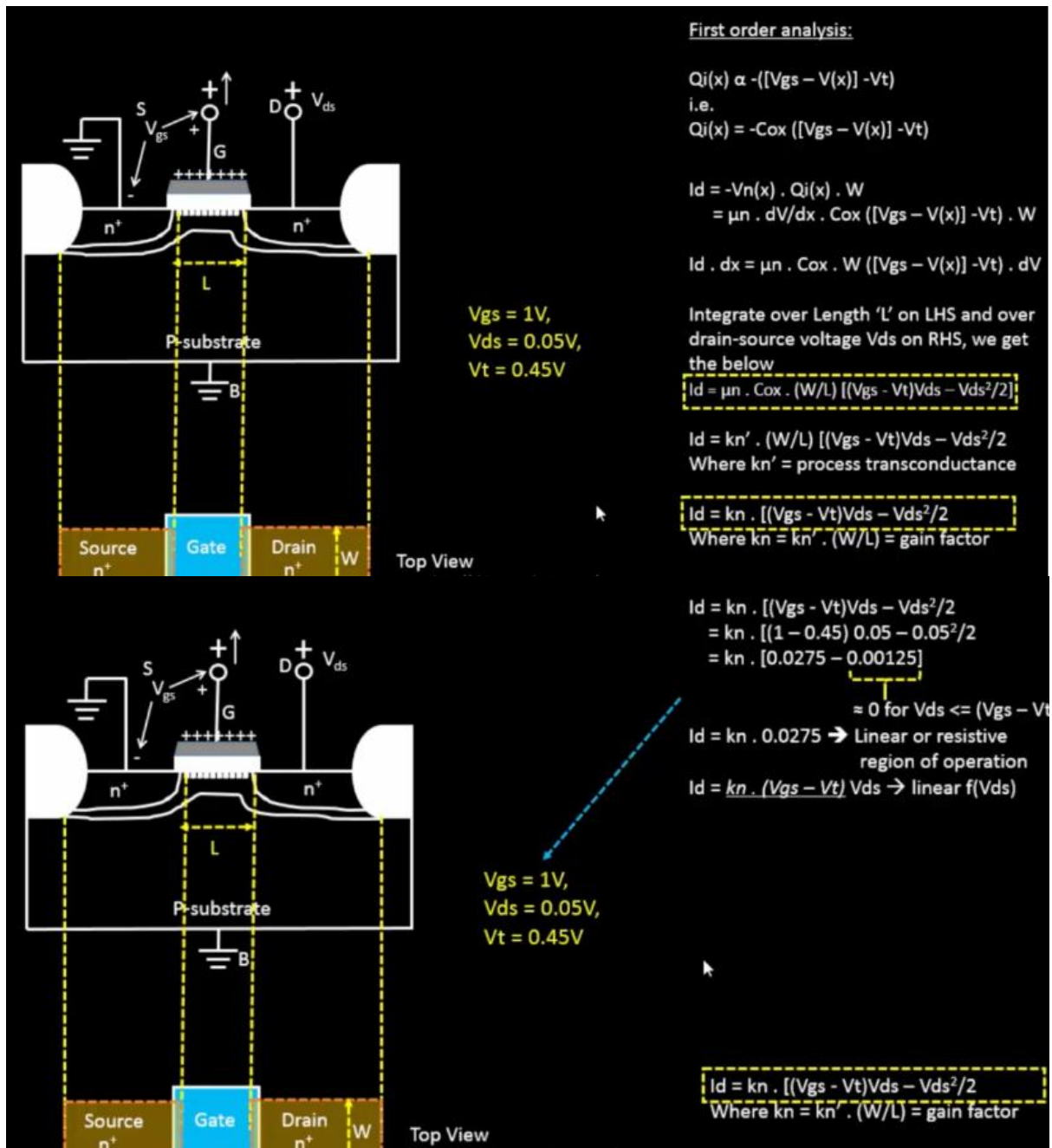
Cox: A constant value provided by the foundry, representing oxide capacitance.

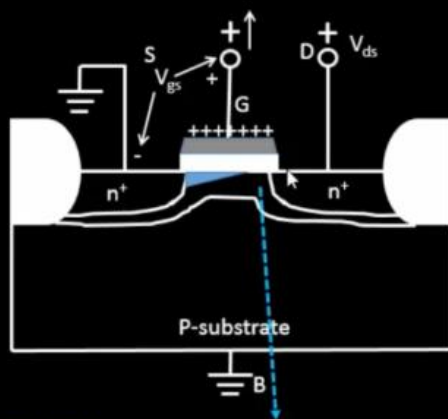




Drain current model for linear region of operation







$V_{gs} = 1V$,
 $V_{ds} = 0.05V$,
 $V_t = 0.45V$

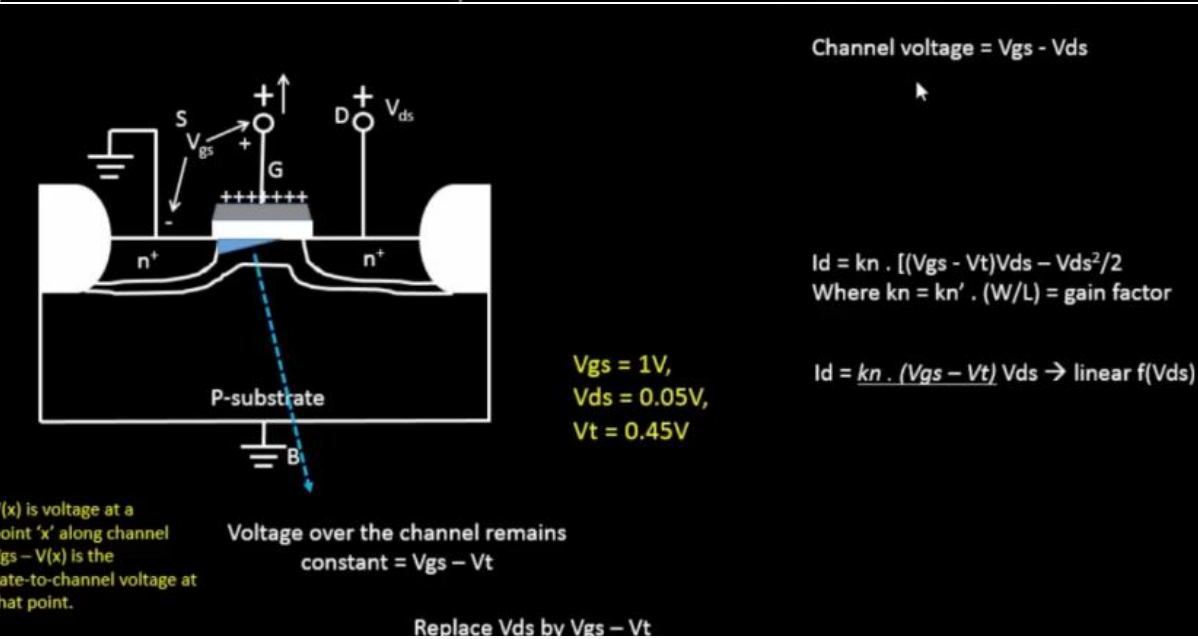
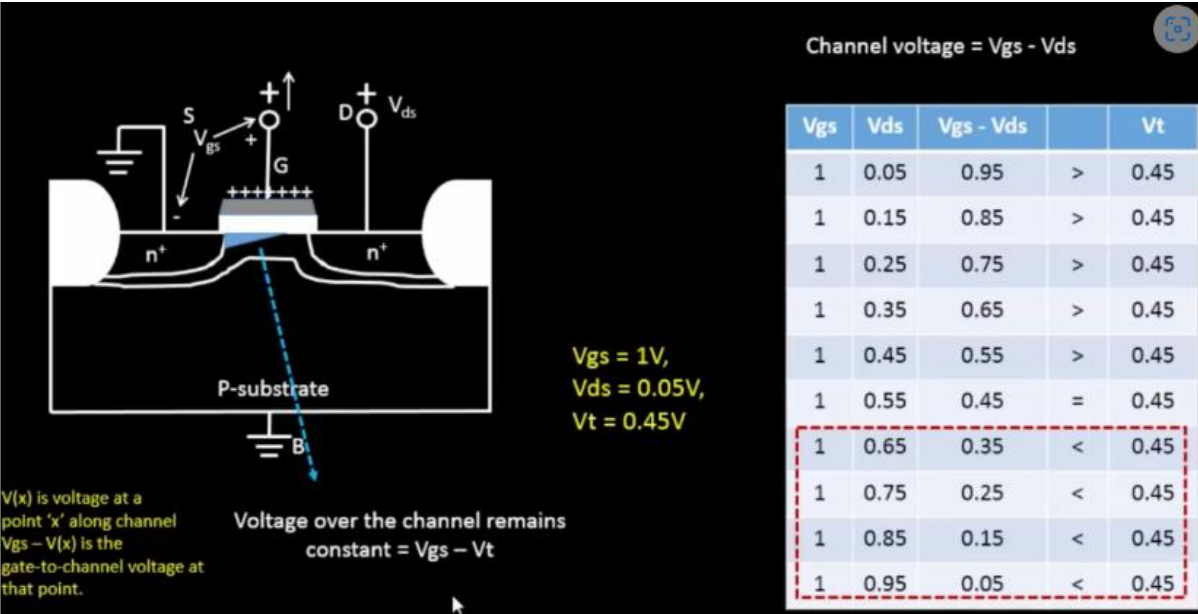
$V(x)$ is voltage at a point 'x' along channel
 $V_{gs} - V(x)$ is the gate-to-channel voltage at that point.

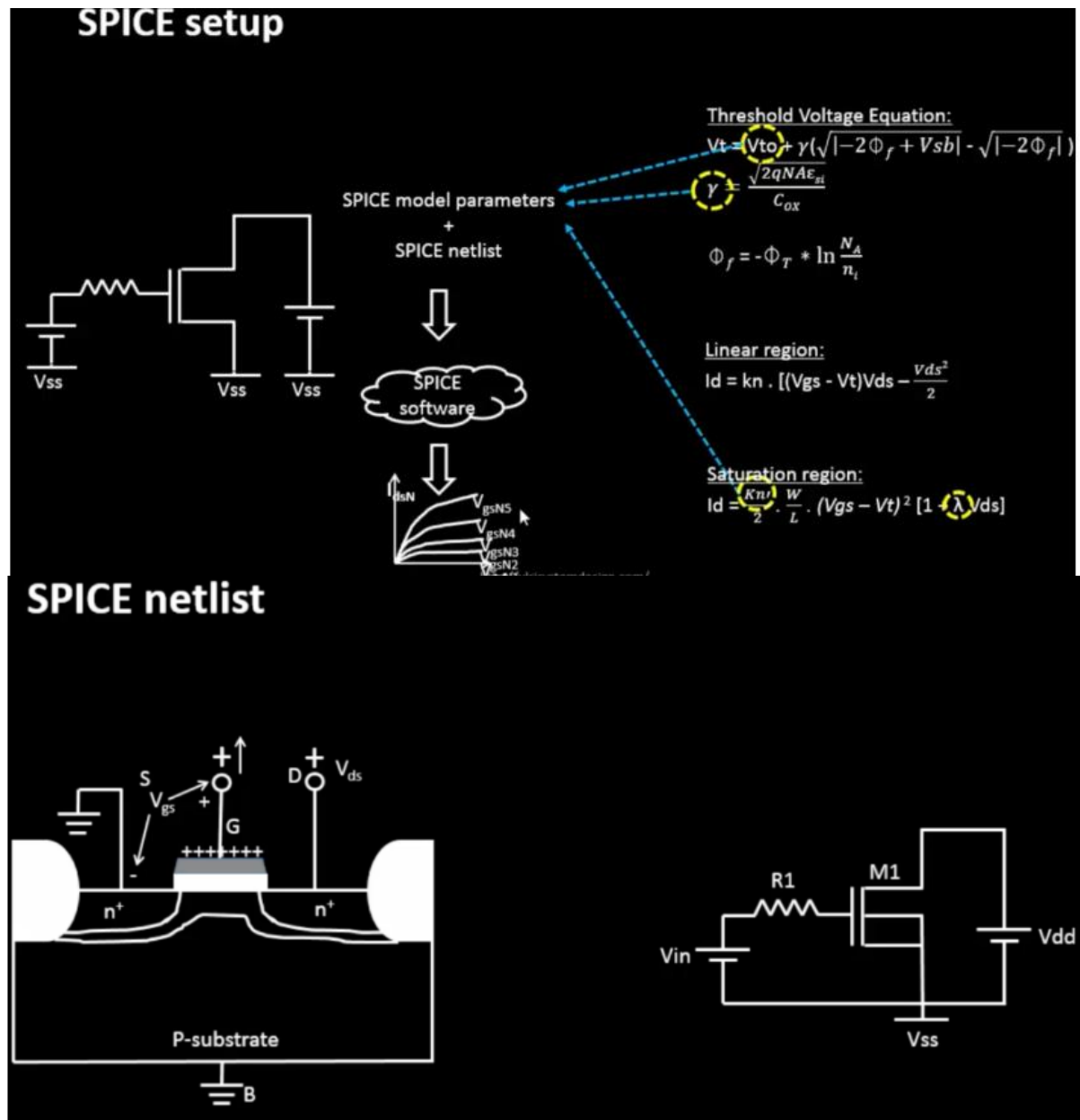
Pinch-Off region – no channel near drain region

Pinch-off condition : $V_{gs} - V_{ds} \leq V_t$

Channel voltage = $V_{gs} - V_{ds}$

V_{gs}	V_{ds}	$V_{gs} - V_{ds}$		V_t
1	0.05	0.95	>	0.45
1	0.15	0.85	>	0.45
1	0.25	0.75	>	0.45
1	0.35	0.65	>	0.45
1	0.45	0.55	>	0.45
1	0.55	0.45	=	0.45
1	0.65	0.35	<	0.45
1	0.75	0.25	<	0.45
1	0.85	0.15	<	0.45
1	0.95	0.05	<	0.45





SPICE Netlist Example:

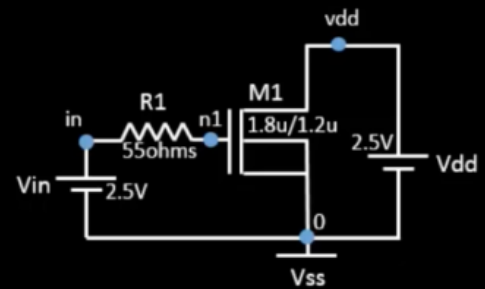
- **M1 vdd n1 0 0 nmos W=1.8u L=1.2u:** Defines an **NMOS** transistor with gate width 1.8μ and length 1.2μ, connecting **vdd** (drain), **n1** (gate), **0** (source and substrate).
- **R1 in n1 55:** Defines a **resistor** with value 55Ω between **in** and **n1**.
- **Vdd vdd 0 2.5:** Defines a **voltage source** of 2.5V between **vdd** and ground (0).
- **Vin in 0 2.5:** Defines an **input voltage source** of 2.5V between **in** and ground (0).

This structure is used to describe circuit components and connections for simulation.

```

M1 vdd n1 0 0 nmos W=1.8u L=1.2u
R1 in n1 55
Vdd vdd 0 2.5
Vin in 0 2.5

```



Model to get NMOS

SPICE netlist Technology file

```

M1 vdd n1 0 0 nmos W=1.8u L=1.2u
R1 in n1 55
Vdd vdd 0 2.5
Vin in 0 2.5

```

```

.lib cmos_models
.MODEL nmos NMOS (TOX = ..
+ VTH0 = .. U0 = .. GAMMA1 = ..)
.MODEL pmos PMOS (TOX = ..
+ VTH0 = .. U0 = .. GAMMA1 = ..)
.endl

```



Package in a file

xxxx_025um_model.mod

Threshold Voltage Equation:

$$V_t = V_{to} + \gamma (\sqrt{|-2\Phi_f + V_{sb}|} - \sqrt{|-2\Phi_f|})$$

$$\gamma = \frac{\sqrt{2qNA\epsilon_{si}}}{C_{ox}}$$

$$\Phi_f = -\Phi_T \ln \frac{N_A}{n_i}$$

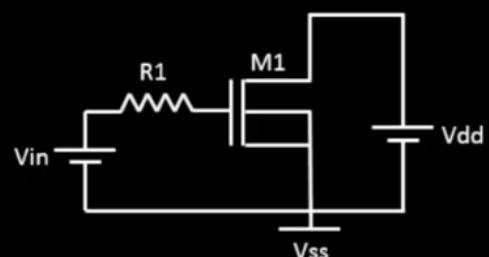
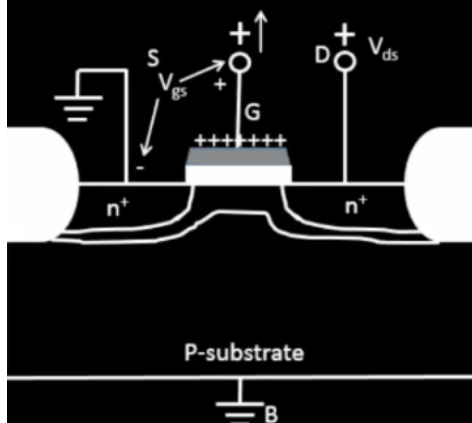
Linear region:

$$I_d = k_n \cdot [(V_{gs} - V_t)V_{ds} - \frac{V_{ds}^2}{2}]$$

Saturation region:

$$I_d = \frac{k_n}{2} \cdot \frac{W}{L} \cdot (V_{gs} - V_t)^2 [1 - \lambda V_{ds}]$$

SPICE netlist



LABS

1. Introduction to SPICE

SPICE (Simulation Program with Integrated Circuit Emphasis) is a powerful tool used for simulating the behavior of electronic circuits. It allows circuit designers to analyze and validate designs before fabrication by providing insights into performance metrics such as voltage, current, and power consumption.

SPICE Lab with Sky130 Models

To use SPICE with Sky130 technology, you can clone the relevant GitHub repository containing Sky130 models and circuits for simulation.

- **Clone the repo:**

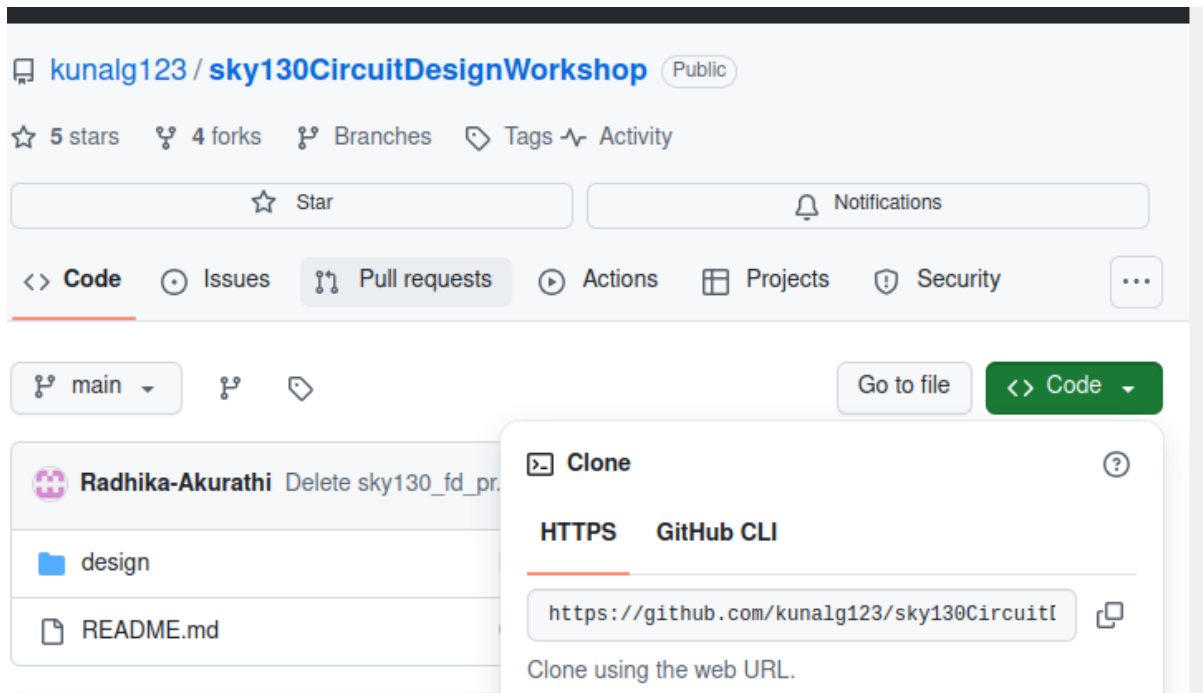
Clone the repository with the following command:

```
git clone https://github.com/kunalg123/sky130CircuitDesignWorkshop.git
```

3 Important Files in the Repo:

1. `/sky130CircuitDesignWorkshop/design/sky130_fd_pr/cells/nfet_01v8/sky130_fd_pr__nfet_01v8__tt.pm3.spice`
This file contains the SPICE model for the **NFET (N-channel MOSFET)** in the Sky130 process at typical (tt) conditions.
2. `/sky130CircuitDesignWorkshop/design/sky130_fd_pr/cells/nfet_01v8/sky130_fd_pr__nfet_01v8__tt.corner.spice`
This file provides the corner model for the **NFET**, used for simulating different process variations.
3. `/sky130CircuitDesignWorkshop/design/sky130_fd_pr/models/sky130.lib.pm3.spice`
This library file contains all the **SPICE models** for components in the Sky130 process node.

You can download ngspice for Windows from the official source using the following link: [ngspice Downloads - SourceForge](#)



```
mjcet@ubuntuvsd:~/Desktop/raheem/week4/sky130CircuitDesignWorkshop/design$ ls
bsim4v5.out          day3_inv_vtc_Wp084_Wn036.spice
day1_nfet_idvds_L2_W5.spice  day4_inv_noisemargin_wp1_wn036.spice
day2_nfet_idvds_L015_W039.spice  day5_inv_devicevariation_wp7_wn042.spice
day2_nfet_idvgs_L015_W039.spice  day5_inv_supplyvariation_Wp1_Wn036.spice
day3_inv_tran_Wp084_Wn036.spice  sky130_fd_pr
mjcet@ubuntuvsd:~/Desktop/raheem/week4/sky130CircuitDesignWorkshop/design$ ngsp
ice day1_nfet_idvds_L2_W5.spice
```

EXAMPLE

```
*Model Description
.param temp=27

*Including sky130 library files
.lib "sky130_fd_pr/models/sky130.lib.spice" tt

*Netlist Description
XM1 Vdd n1 0 0 sky130_fd_pr__nfet_01v8 w=5 l=2
R1 n1 in 55
Vdd vdd 0 1.8V
Vin in 0 1.8V

*simulation commands
.op
.dc Vdd 0 1.8 0.1 Vin 0 1.8 0.2

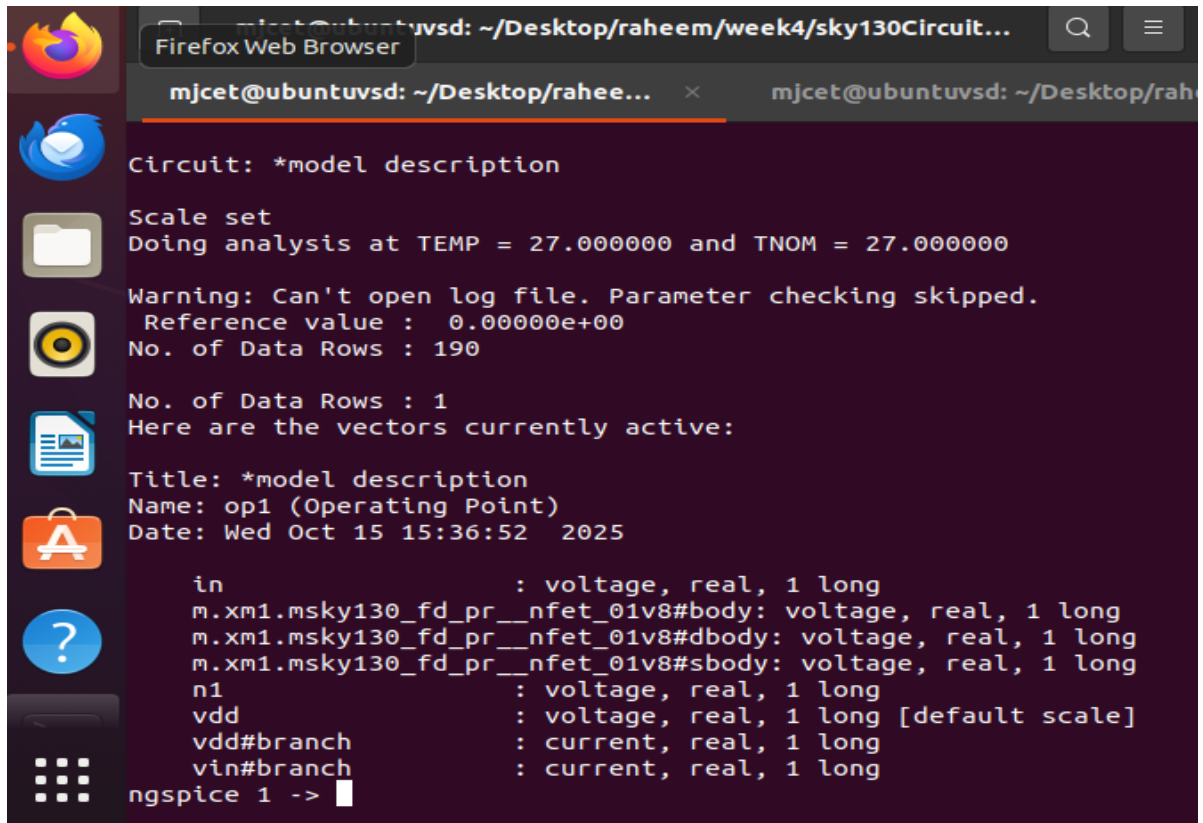
.control

run
display
setplot dc1
.endc
.end
```

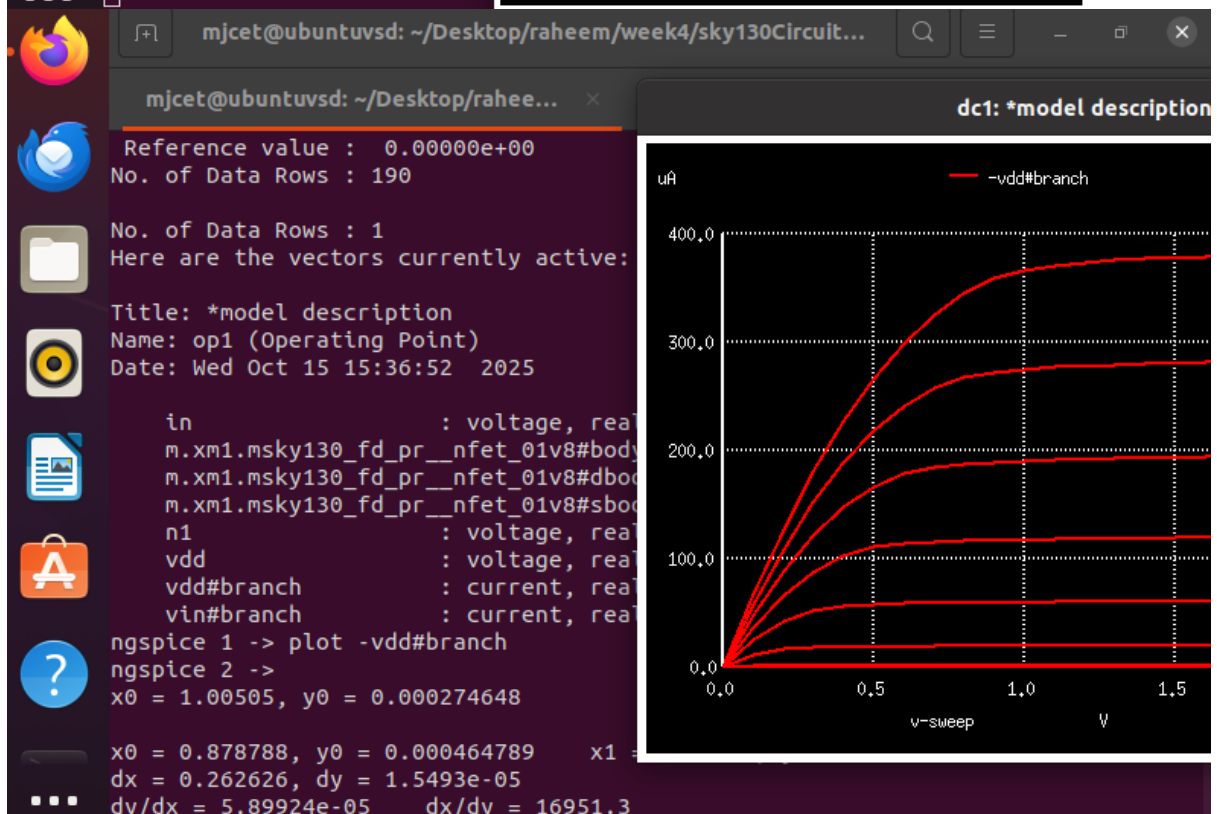
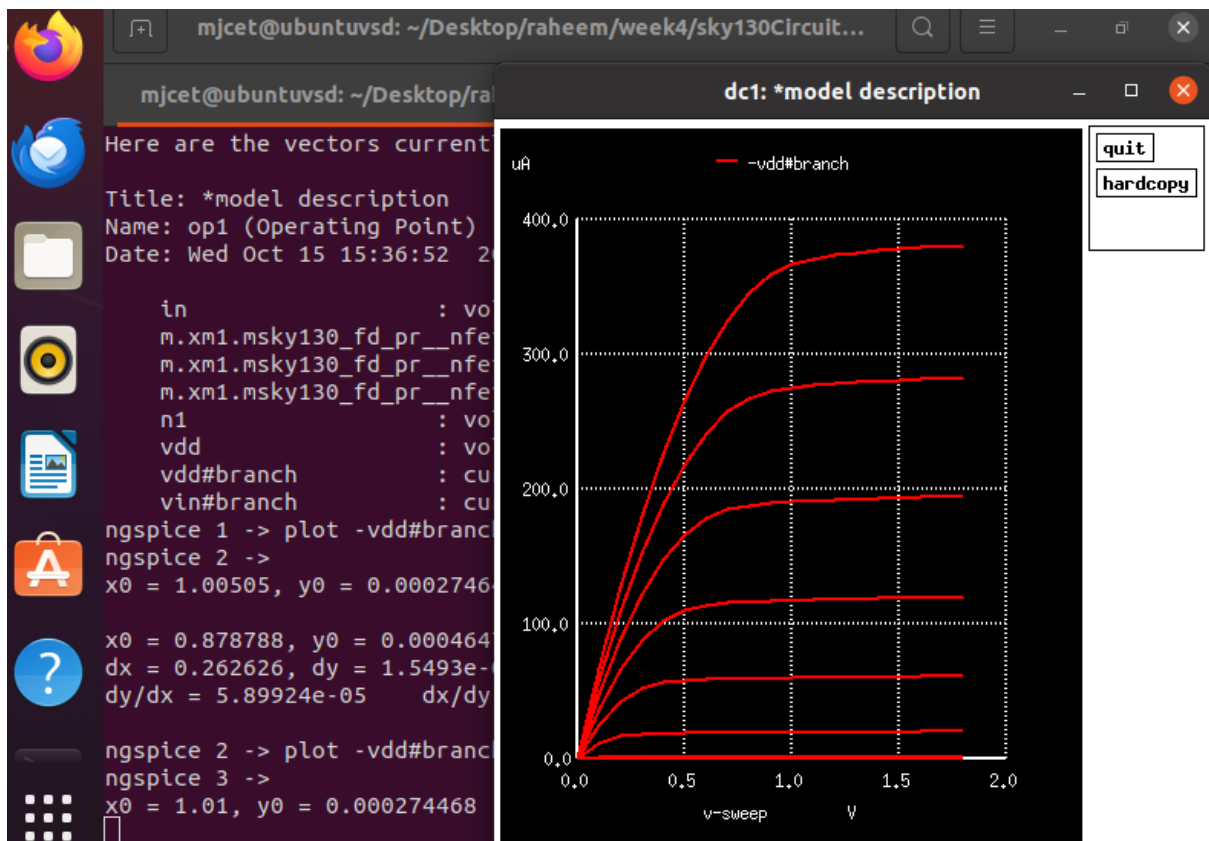
To plot the waveforms in ngspice

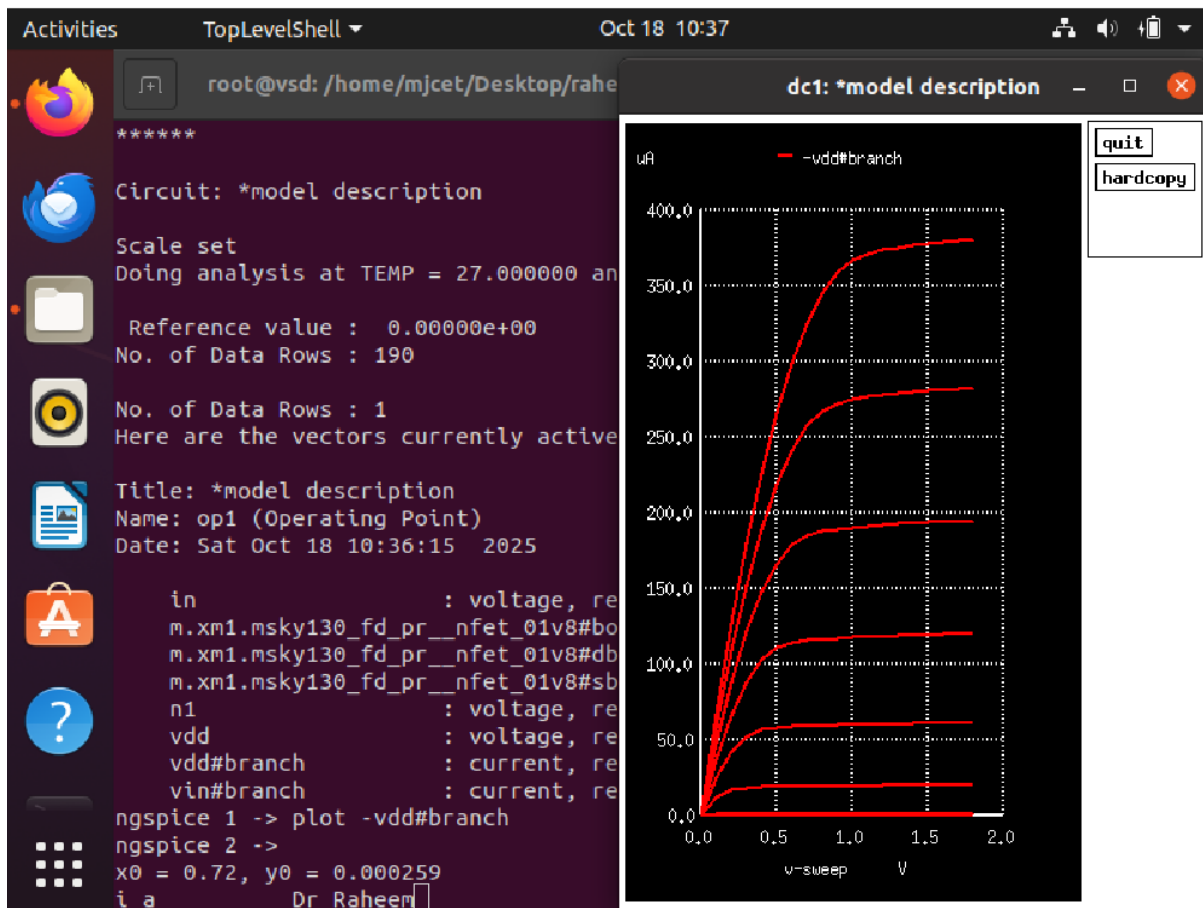
```
ngspice day1_nfet_idvds_L2_W5.spice  
plot -vdd#branch
```

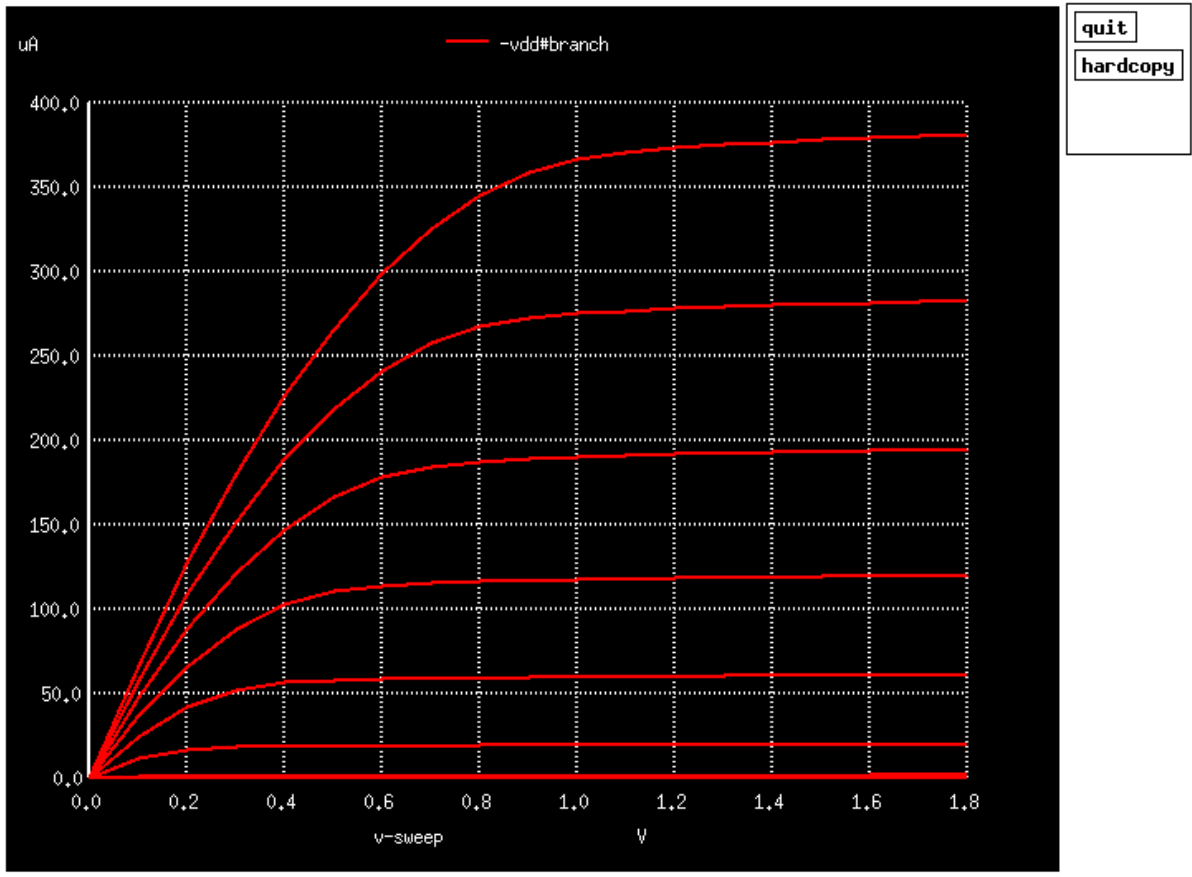
The plot of I_{ds} vs V_{ds} over constant V_{gs} :



```
mjcet@ubuntuvsd: ~/Desktop/raheem/week4/sky130Circuit...  
Circuit: *model description  
Scale set  
Doing analysis at TEMP = 27.000000 and TNOM = 27.000000  
Warning: Can't open log file. Parameter checking skipped.  
Reference value : 0.000000e+00  
No. of Data Rows : 190  
No. of Data Rows : 1  
Here are the vectors currently active:  
Title: *model description  
Name: op1 (Operating Point)  
Date: Wed Oct 15 15:36:52 2025  
in : voltage, real, 1 long  
m.xm1.msky130_fd_pr__nfet_01v8#body: voltage, real, 1 long  
m.xm1.msky130_fd_pr__nfet_01v8#dbody: voltage, real, 1 long  
m.xm1.msky130_fd_pr__nfet_01v8#sbody: voltage, real, 1 long  
n1 : voltage, real, 1 long  
vdd : voltage, real, 1 long [default scale]  
vdd#branch : current, real, 1 long  
vin#branch : current, real, 1 long  
ngspice 1 ->
```







Thank you VSD Team for mentoring the RSIC V Reference SoC Tapeout Program

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